CONVERGENCE AND FINITE LENGTH OF THE BRASCAMP–LIEB SCALING FLOW

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Abstract. We give a proof of full convergence of the prescribed Brascamp–Lieb scaling flow for every feasible real datum with positive real weights, answering Question 2.10 of Bez, Gauvan, and Tsuji. The limit is geometric, and the sequence of complete scaling iterations has finite length in the product Frobenius norm. Stacking the matrices with square-root weights identifies the two normalizations with nearest-point projections onto a product of scaled row-Stiefel manifolds and a column-Stiefel manifold. The result then follows from the asymptotic normalization theorem of Bez, Gauvan, and Tsuji and the established local alternating-projection theory of Noll and Rondepierre. We also give a direct verification of the descent conditions of Attouch, Bolte, and Svaiter. The one-dimensional-output recurrence agrees with the prescribed-norm frame algorithm studied by Zhu and Li. An explicit feasible example has infinite interleaved half-step length and error of order $k^{-1/2}$, clarifying the scope of the complete-iteration conclusion.

1. Introduction and main result

The Brascamp–Lieb scaling iteration alternately normalizes the rows within each matrix block and the weighted sum of the blocks' Gram matrices. Each step enforces one identity while generally disturbing the other. Asymptotic satisfaction of both identities does not by itself imply convergence of the matrices: a sequence may approach the constraint set while continuing to move along it. Bez, Gauvan, and Tsuji [1] established asymptotic normalization and convergence along a subsequence. Their Question 2.10 asks whether the whole scaling sequence converges. We prove convergence of that sequence and summability of its movement over complete iterations.

Let $n, n_1, \dots, n_m$ be positive integers, let $c_j > 0$, and let $B_j : \mathbb{R}^n \to \mathbb{R}^{n_j}$ be linear maps. The Brascamp–Lieb constant $\mathrm{BL}(\mathbf{B}, \mathbf{c})$ is the least constant $C \in [0, \infty]$ such that

$$\int_{\mathbb{R}^n} \prod_{j=1}^{m} f_j(B_j x)^{c_j}\, dx \le C \prod_{j=1}^{m} \left(\int_{\mathbb{R}^{n_j}} f_j \right)^{c_j} \tag{1.1}$$

for all nonnegative integrable functions f_j. The datum is *feasible* when this constant is finite. Throughout, $*$ denotes transpose. For tuples and matrices we use the Frobenius norms

$$\|A\|_F^2 = \mathrm{tr}(A^* A), \qquad \|\mathbf{B}\|_F^2 = \sum_{j=1}^{m} \|B_j\|_F^2 .$$

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Define the two normalizations by

$$\begin{aligned} S(\mathbf{B}) &= \sum_{j=1}^{m} c_j B_j^* B_j, \qquad \psi(\mathbf{B})_j = B_j S(\mathbf{B})^{-1/2}, \\ \phi(\mathbf{B})_j &= (B_j B_j^*)^{-1/2} B_j, \qquad \Phi = \phi \circ \psi. \end{aligned} \tag{1.2}$$

Feasibility implies that the blocks are surjective, their common kernel is zero, and $\sum_j c_j n_j = n$; see [1, 2]. Invertible changes of coordinates in the domain and codomains preserve feasibility. Consequently both normalizations are well defined and preserve feasibility at every finite step. The map ψ enforces $S(\psi(\mathbf{B})) = I_n$, while ϕ enforces orthonormal rows within every block. A datum is called *geometric* when both identities hold. We call $\Phi = \phi \circ \psi$ one complete iteration; $\psi(\mathbf{B})$ is the intermediate half-step.

Theorem 1.1. *Let $(\mathbf{B}^0, \mathbf{c})$ be a feasible real Brascamp–Lieb datum with $c_j > 0$ and*

$$B_j^0 (B_j^0)^* = I_{n_j} \qquad (1 \le j \le m).$$

Set $\mathbf{B}^{k+1} = \Phi(\mathbf{B}^k)$ for $k \ge 0$. Then there is a tuple $\mathbf{G}$ such that

$$\mathbf{B}^k \longrightarrow \mathbf{G}, \qquad G_j G_j^* = I_{n_j}, \qquad \sum_{j=1}^{m} c_j G_j^* G_j = I_n. \tag{1.3}$$

Moreover, the complete iterates have finite length:

$$\sum_{k=0}^{\infty} \left\| \mathbf{B}^{k+1} - \mathbf{B}^k \right\|_F < \infty. \tag{1.4}$$

The weights need not be rational, and no extremisability assumption is imposed. A feasible starting tuple without the initial row normalization enters the stated domain after one complete Φ-step, so the conclusions hold for that trajectory as well. The limit may belong to the closure of the equivalence orbit without belonging to the orbit itself. The theorem concerns the normalized matrices; it does not assert convergence of the accumulated changes of coordinates or an effective uniform convergence rate.

The recurrence comes from the scaling algorithm of Garg, Gurvits, Oliveira, and Wigderson [2], in the normalization order used in [1]. Convergence to a geometric datum is already described in the introduction of [2, p. 107]; their Theorem 10.3 gives a finite-accuracy bound for the geometricity residual. Earlier, Gurvits [3, Theorem 4.6] established vanishing double-stochasticity residual for rank-nondecreasing positive operators. The purpose here is to give an explicit argument for convergence of the prescribed matrix tuples and finite length of their complete iterations, while acknowledging the earlier convergence assertion.

The argument retains the entire tuple in weighted matrix coordinates. In those coordinates the two normalizations are alternating Euclidean projections between compact algebraic manifolds. The local theorem of Noll and Rondepierre [4], including the finite-length estimate in its proof, then applies at a geometric cluster point supplied by [1]. Appendix A gives a direct verification through Attouch, Bolte, and Svaiter [5], with uniform polar-decomposition estimates. These are applications of existing convergence principles.

There is also a direct predecessor in frame theory. Tropp, Dhillon, Heath, and Strohmer [6] studied alternating projection for structured tight frames. Zhu and Li [7, Theorem 6] state full convergence for prescribed column norms; finite component length also appears in the proof of their general theorem. When every output dimension is one, transposition of our weighted stack

gives exactly that recurrence. We spell out the correspondence in Section 2. The matrix-block formulation additionally retains orthogonality among the rows within each output block.

Recent geometric approximation results address related questions. Watanabe [8] constructs approximately geometric representatives and obtains geometric limits by subsequence extraction; Bez, Gauvan, and Tsuji [9] study regularity of operator and quiver capacity; and Chindris and Derksen [10, Corollary 14] recover geometric ubiquity through a selected degeneration, with rational positive weights in the cited revision. These conclusions concern existence, capacity, or selected representatives. Here the trajectory is fixed by the normalization recurrence. Likewise, the operator-scaling convergence results of Soma and Uschmajew [11] use a positivity-improving hypothesis, and their Proposition 2.3 relates factor and tuple recurrences through potentially time-dependent orthogonal transformations. We work directly with the prescribed tuples in fixed coordinates.

2. The projection formulation

Put $d = \sum_j n_j$ and identify a tuple with a matrix $X \in \mathbb{R}^{d\times n}$ by stacking the weighted blocks

$$X_j = \sqrt{c_j}\, B_j \qquad (1 \le j \le m). \tag{2.1}$$

This is an invertible linear change of coordinates. The two normalization constraints become

$$\begin{aligned} M &= \{X \in \mathbb{R}^{d\times n} : X_j X_j^* = c_j I_{n_j} \text{ for every } j\}, \\ N &= \{Y \in \mathbb{R}^{d\times n} : Y^*Y = I_n\}. \end{aligned} \tag{2.2}$$

Here M is a product of scaled row-Stiefel manifolds and N is a column-Stiefel manifold. They are compact smooth real algebraic manifolds. The required inequalities $n_j \le n$ and $d \ge n$ follow respectively from block surjectivity and the trivial common kernel. Their intersection corresponds exactly to the geometric identities in (1.3). Thus the row constraints and the weighted Gram constraint become two sets in one Euclidean space. The weights are absorbed into the coordinates, so both normalizations can be interpreted using the same Frobenius distance.

The projection formulas and the quantitative estimate needed below both follow from the same elementary identity.

Lemma 2.1. *Let A have full column rank and write its polar decomposition as $A = QH$, where $Q^*Q = I$ and $H = (A^*A)^{1/2} > 0$. For every W of the same size with $W^*W = I$,*

$$\frac{1}{2}\,\|A - W\|_F^2 - \frac{1}{2}\,\|A - Q\|_F^2 = \frac{1}{2}\,\mathrm{tr}\big(H(W-Q)^*(W-Q)\big) \ge \frac{\sigma_{\min}(A)}{2}\,\|W - Q\|_F^2\,. \tag{2.3}$$

In particular, Q is the unique Frobenius nearest point to A among matrices with orthonormal columns.

Proof. Since $W^*W = Q^*Q = I$, the difference on the left equals

$$\mathrm{tr}(H) - \mathrm{tr}(HQ^*W).$$

Expanding $(W-Q)^*(W-Q)$ gives the equality in (2.3); the two cross terms have equal trace because H is symmetric. The inequality follows from $H \ge \sigma_{\min}(A) I$. Since $\sigma_{\min}(A) > 0$, equality of the two distances forces $W = Q$. □

Thus, wherever the indicated ranks are full, the nearest-point projections onto (2.2) are

$$P_N X = X(X^*X)^{-1/2}, \qquad (P_M Y)_j = \sqrt{c_j}(Y_j Y_j^*)^{-1/2} Y_j. \tag{2.4}$$

The second formula is obtained by transposing each block and projecting onto matrices with orthonormal columns, then multiplying by $\sqrt{c_j}$.

Let $X_j^k = \sqrt{c_j} B_j^k$ and $Y^k = P_N X^k$. Substitution in (1.2) and (2.4) gives

$$X^{k+1} = P_M Y^k, \qquad Y^{k+1} = P_N X^{k+1}. \tag{2.5}$$

All these projections are unique by feasibility and Lemma 2.1. The identification is exact in the fixed Frobenius metric; no change by an iteration-dependent orthogonal matrix is involved.

When $n_j = 1$ for every j, set $V = X^*$, whose jth column is $\sqrt{c_j} B_j^*$. The constraint M prescribes squared column norms c_j, while N becomes the tight-frame constraint $VV^* = I_n$. Since $\sum_j c_j = n$, the tightness parameter of [7, Section 3.1] is $a = 1$. Transposition carries both projections in (2.4) to their frame projections. Feasibility supplies the full rank and nonzero columns required in their Theorem 6. This identifies the scalar-output recurrence with their stated frame convergence result. The proof below establishes both the scalar and matrix-block conclusions independently of that argument, using BL residual convergence to identify a geometric limit. For larger output blocks, independent column normalization of X^* would omit the within-block orthogonality constraints in M.

The asymptotic input from Brascamp–Lieb scaling theory is Proposition 2.8 of [1], which gives

$$(X^k)^* X^k \longrightarrow I_n. \tag{2.6}$$

Since $X^k \in M$ is bounded, continuity of the inverse square root near I_n and (2.4) imply

$$Y^k - X^k \longrightarrow 0. \tag{2.7}$$

3. Convergence and finite length

We now have a bounded alternating-projection sequence whose two components approach one another. This gives a common cluster point in $M \cap N$, but compactness alone does not control the rest of the trajectory. The local theorem of Noll and Rondepierre [4, Corollaries 3 and 8] provides that control. For the subanalytic sets used here, with one set locally prox-regular, alternating projections started sufficiently near a common point converge to an intersection point. The proof of their Theorem 1 also gives summability of the successive increments of the component on the regular set. In our application that component is X^k on M.

Both hypotheses have concrete sources here. The sets are real algebraic, hence subanalytic. Their smooth manifold structure gives local prox-regularity; near an intersection point, the full-rank polar formulas also exhibit the unique smooth nearest-point projections. It remains to show that the prescribed trajectory enters the neighborhood where the local result applies.

Proof of Theorem 1.1. Compactness of M gives a subsequence $X^{k_\ell} \to X^\star$. Equation (2.7) gives $Y^{k_\ell} \to X^\star$ as well. Since M and N are closed, $X^\star \in M \cap N$.

Apply the local theorem with its sets $A = N$ and $B = M$. They are closed subanalytic sets, and M is locally prox-regular at $X^\star$. The projection order (2.5) matches the theorem, with its B-component equal to X^k after an index shift. The common subsequence enters the required neighborhood, so the full sequence converges to $X^\star$. The summability estimate in the proof of [4, Theorem 1] gives

$$\sum_{k=0}^{\infty} \left\| X^{k+1} - X^k \right\|_F < \infty, \tag{3.1}$$

since adding finitely many initial increments does not change finiteness.

Set $G_j = X_j^\star / \sqrt{c_j}$. Membership in $M \cap N$ gives the geometric identities (1.3). With $c_{\min} = \min_j c_j > 0$, the fixed change of coordinates satisfies

$$\left\| \mathbf{B}^{k+1} - \mathbf{B}^k \right\|_F \le c_{\min}^{-1/2} \left\| X^{k+1} - X^k \right\|_F .$$

Equation (3.1) therefore proves (1.4). □

The conclusion supplies a single limiting tuple in the original coordinates and summable cumulative movement over complete iterations. It strengthens residual control in this specific sense. It does not by itself give a computable stopping criterion or a uniform linear rate. The distinction between complete iterations and the intervening normalizations is necessary, as the following example shows.

4. An example with infinite half-step length

Finite length in Theorem 1.1 refers to complete iterations. It does not assert finite length of the path that visits each normalization half-step. Zigzagging of alternating normalization steps is discussed in [12, Sections 3.1–3.2]. The following exact example distinguishes the two lengths and also prevents a linear convergence claim for all feasible data.

Take $n = 2$, $n_1 = n_2 = n_3 = 1$, $\mathbf{c} = (1, \frac{1}{2}, \frac{1}{2})$, and

$$B_1 = (1, 0), \qquad B_2 = (a, \sqrt{1-a^2}), \qquad B_3 = (a, -\sqrt{1-a^2}), \qquad 0 < a < 1. \tag{4.1}$$

The rows are normalized. To check feasibility directly, put $b = \sqrt{1-a^2}$. Cauchy–Schwarz in y, followed by integration in x, gives

$$\begin{aligned}\int_{\mathbb{R}^2} f_1(x) f_2(ax+by)^{1/2} f_3(ax-by)^{1/2}\,dx\,dy \\ \le b^{-1}\left(\int_{\mathbb{R}} f_1\right)\left(\int_{\mathbb{R}} f_2\right)^{1/2}\left(\int_{\mathbb{R}} f_3\right)^{1/2}.\end{aligned}$$

Thus (4.1) is feasible for every $0 < a < 1$.

Here $S(\mathbf{B}) = \operatorname{diag}(1+a^2, 1-a^2)$. The ψ-step gives the rows

$$((1+a^2)^{-1/2}, 0), \qquad \left(\frac{a}{\sqrt{1+a^2}}, 1\right), \qquad \left(\frac{a}{\sqrt{1+a^2}}, -1\right).$$

Normalizing their lengths returns the form (4.1), with parameter

$$a_{k+1} = \frac{a_k}{\sqrt{1+2a_k^2}}, \qquad a_k^2 = \frac{1}{a_0^{-2} + 2k}. \tag{4.2}$$

The second equality follows by taking reciprocal squares in the first. Consequently the limit is

$$G_1 = (1, 0), \qquad G_2 = (0, 1), \qquad G_3 = (0, -1).$$

The first row alone supplies the half-step lower bound

$$\left\|\psi(\mathbf{B}^k) - \mathbf{B}^k\right\|_F \ge 1 - (1+a_k^2)^{-1/2} \sim \frac{1}{4k}. \tag{4.3}$$

In this example the first row is $(1, 0)$ at every complete iterate. The ψ-step shortens it, and the following ϕ-step restores it exactly. The lower bound in (4.3) therefore records movement that disappears when only complete iterates are compared.

The total length of the interleaved path

$$\mathbf{B}^0,\ \psi(\mathbf{B}^0),\ \mathbf{B}^1,\ \psi(\mathbf{B}^1),\ \mathbf{B}^2, \ldots$$

therefore diverges. The same is true in the weighted stacked coordinates, because $c_1 = 1$. This does not conflict with (1.4): its increments compare points on the same constraint set at successive complete iterations. The paired-sequence estimate in Appendix A has the same interpretation.

Finally,

$$\left\|\mathbf{B}^k - \mathbf{G}\right\|_F^2 = 2(a_k^2 + (\sqrt{1-a_k^2} - 1)^2) \sim 2a_k^2 \sim k^{-1}.$$

Thus the error is asymptotic to $k^{-1/2}$ and cannot satisfy a geometric bound Cq^k with $0 < q < 1$. The convergence and finite-length assertions hold without such a bound.

Appendix A. Direct descent verification

For completeness, we verify the descent conditions directly in the block representation. This gives a second route to convergence and finite length of the paired complete iterates. The squared distance between the current projected points decreases at each projection, but a decrease estimate alone would control only the sum of squared movements. A bound on the subgradient in terms of the next movement allows the Kurdyka–Łojasiewicz argument to give the sum of the movements themselves. We use the descent theorem of Attouch, Bolte, and Svaiter [5, Theorem 2.9]. In the form needed here, it states the following. Let F be a proper lower semicontinuous function with the Kurdyka–Łojasiewicz property, and let (z^k) satisfy, for constants $a, b > 0$,

$$F(z^{k+1}) + a\left\|z^{k+1} - z^k\right\|_F^2 \le F(z^k), \tag{A.1}$$

$$w^{k+1} \in \partial F(z^{k+1}), \qquad \left\|w^{k+1}\right\|_F \le b\left\|z^{k+1} - z^k\right\|_F. \tag{A.2}$$

If a subsequence z^{k_ℓ} converges to $\bar{z}$ with $F(z^{k_\ell}) \to F(\bar{z})$, then the full sequence converges and has finite length. Here ∂ denotes the limiting subdifferential. Proper lower semicontinuous semialgebraic functions have the required Kurdyka–Łojasiewicz property, as recalled in [5].

For the alternating projections, use the paired sequence $z^k = (X^k, Y^k)$ with product norm

$$\|(U, V)\|_F^2 = \|U\|_F^2 + \|V\|_F^2,$$

and define

$$F(X, Y) = \frac{1}{2}\|X - Y\|_F^2 + \iota_M(X) + \iota_N(Y), \tag{A.3}$$

where ι_C is zero on C and $+\infty$ elsewhere. The points $z^k = (X^k, Y^k)$ record a pair after each complete iteration. Their increments compare X^{k+1} with X^k and Y^{k+1} with Y^k; they do not measure the distances from X^k to Y^k within a pair. This function is proper, lower semicontinuous, and semialgebraic. The last assertion holds for arbitrary positive real c_j: a semialgebraic description permits real coefficients, and the constraints in (2.2) are polynomial equations.

Write $c_{\min} = \min_j c_j > 0$. The descent coefficients can be bounded uniformly along the trajectory, with constants depending on n and the fixed weights, without first invoking asymptotic normalization. On M, $\left\|X^k\right\|_F^2 = n$ and $X_j^k(X_j^k)^* = c_j I_{n_j}$. Since $Y_j^k = X_j^k((X^k)^* X^k)^{-1/2}$, we have

$$Y_j^k(Y_j^k)^* = X_j^k((X^k)^* X^k)^{-1}(X_j^k)^* \succeq \frac{c_j}{\|X^k\|_{\mathrm{op}}^2} I_{n_j}.$$

Consequently

$$\sigma_{\min}(Y_j^k) \ge \frac{\sqrt{c_j}}{\|X^k\|_{\mathrm{op}}} \ge \sqrt{c_j/n}. \tag{A.4}$$

Here the least singular value of a full-row-rank block is the square root of the least eigenvalue of its row Gram matrix. Since $(Y^k)^* Y^k = I_n$, each $Y_j^k(Y_j^k)^* \preceq I_{n_j}$. All these row Gram matrices are invertible by rank preservation. Thus

$$\begin{aligned}(X^{k+1})^* X^{k+1} &= \sum_j c_j (Y_j^k)^* \left(Y_j^k (Y_j^k)^*\right)^{-1} Y_j^k \\ &\succeq c_{\min} \sum_j (Y_j^k)^* Y_j^k = c_{\min} I_n.\end{aligned} \tag{A.5}$$

Direct verification of convergence and finite length. We verify (A.1)–(A.2) for every $k \geq 0$. First split the decrease in F into the decreases from the two projections:

$$F(z^k) - F(z^{k+1}) = \frac{1}{2}\left(\left\|X^k - Y^k\right\|_F^2 - \left\|X^{k+1} - Y^k\right\|_F^2\right) + \frac{1}{2}\left(\left\|X^{k+1} - Y^k\right\|_F^2 - \left\|X^{k+1} - Y^{k+1}\right\|_F^2\right).$$

For the first difference, apply Lemma 2.1 to $(Y_j^k)^*/\sqrt{c_j}$ and the comparison matrix $(X_j^k)^*/\sqrt{c_j}$, then multiply by c_j. The resulting lower bound for the jth block is

$$\frac{\sigma_{\min}(Y_j^k)}{2\sqrt{c_j}} \left\|X_j^{k+1} - X_j^k\right\|_F^2.$$

Apply the lemma directly to X^{k+1} for the second difference. Summing the block estimates yields

$$\begin{aligned} F(z^k) - F(z^{k+1}) \geq \frac{1}{2} \min_j \frac{\sigma_{\min}(Y_j^k)}{\sqrt{c_j}} \left\|X^{k+1} - X^k\right\|_F^2 \\ + \frac{1}{2}\sigma_{\min}(X^{k+1}) \left\|Y^{k+1} - Y^k\right\|_F^2. \end{aligned} \tag{A.6}$$

Equations (A.4)–(A.5) give (A.1) with the uniform positive constant

$$a = \frac{1}{2}\min\{n^{-1/2}, \sqrt{c_{\min}}\}.$$

For the relative-error condition, write N_M and N_N for the normal spaces, which here agree with the limiting normal cones. Projection optimality gives

$$Y^k - X^{k+1} \in N_M(X^{k+1}), \qquad X^{k+1} - Y^{k+1} \in N_N(Y^{k+1}).$$

The subdifferential of (A.3) is the gradient of its smooth term plus the product of these normal spaces. Adding the displayed normals to the two gradient components at z^{k+1} therefore gives

$$w^{k+1} = (Y^k - Y^{k+1}, 0) \in \partial F(z^{k+1}). \tag{A.7}$$

Since $\left\|w^{k+1}\right\|_F \leq \left\|z^{k+1} - z^k\right\|_F$, condition (A.2) holds with $b = 1$.

Finally, z^k lies in the compact set $M \times N$, so it has a convergent subsequence. On this set, F equals the continuous function $\frac{1}{2}\|X - Y\|_F^2$, giving the required convergence of function values along that subsequence. Applying [5, Theorem 2.9] proves convergence of (z^k) and

$$\sum_{k=0}^{\infty} \left\|z^{k+1} - z^k\right\|_F < \infty. \tag{A.8}$$

This proves convergence to a pair in $M \times N$. To identify a geometric datum, we still use the BL asymptotic normalization input: by (2.7), the limit has the form (X^∞, X^∞), and closedness gives $X^\infty \in M \cap N$.

Set $G_j = X_j^\infty/\sqrt{c_j}$. Membership in $M \cap N$ gives (1.3). Moreover,

$$\left\|\mathbf{B}^{k+1} - \mathbf{B}^k\right\|_F \leq (\min_j c_j)^{-1/2} \left\|X^{k+1} - X^k\right\|_F \leq (\min_j c_j)^{-1/2} \left\|z^{k+1} - z^k\right\|_F.$$

Summing this estimate and using (A.8) proves (1.4). □

AI assistance

AI assisted the development of the weighted projection formulation and proof, the examination of related literature, and the preparation of the manuscript. Responsibility for the mathematical claims, references, and final text rests with the author.

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